



ZIMBABWE

MINISTRY OF HIGHER AND TERTIARY EDUCATION,
INNOVATION, SCIENCE AND TECHNOLOGY
DEVELOPMENT

HIGHER EDUCATION EXAMINATIONS COUNCIL
(HEXCO)

NATIONAL CERTIFICATE

IN

AUTOMOTIVE ENGINEERING

SUBJECT: Engineering Mathematics PAPER NO: 346/1S/S0S
346/1S/S0S

DURATION: 3 Hours

JULY/AUGUST 2021 EXAMINATION

REQUIREMENTS

1. Answer booklet.
2. Non - programmable scientific calculator.

INSTRUCTIONS TO CANDIDATE

1. Section A: Answer ALL questions. (40 marks)
2. Section B: Answer ANY FIVE questions (60 marks)

SECTION A**QUESTION 1**

- (a) Convert $125^{\circ} 47'$ to radians. (1 mark)
- (b) Convert $3,472 \times 10^8$ to preferred standard form. (1 mark)
- (c) Simply $(4pq)^2 \times 5$. (2 marks)
- (d) Evaluate $\log_{16} 8$. (2 marks)
- (e) Evaluate $\ln 6,439$. (1 mark)
- (f) If $y = (x - 3)(x^2 + 7x - 1)$ find $\frac{dy}{dx}$. (3 marks)
- (g) $\int_0^1 (x^3 - 3x^2 + 2x) dx$ (4 marks)
- (h) Determine the diameter of a circle whose circumference is 178,4cm. (3 marks)
- (i) Evaluate B correct to 3 significant figures when $w = 7,20$, $V = 10$ and $g = 9.81$.
 Given $B = \frac{wV^2}{2g}$. (3 marks)
- (j) One side of a right angled triangle is 4cm, the hypotenuse is 5cm. Calculate the missing side. (2 marks)
- (k) Differentiate $y = 3\sqrt{x}$. (3 marks)
- (l) Determine $\int \left(4 + \frac{3x}{7} - 6x^2\right) dx$ (4 marks)
- (m) If $A = \begin{bmatrix} 1 & 4 \\ -3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 5 & -1 \\ 0 & 1 \end{bmatrix}$
 Find $A + B$. (4 marks)
- (n) Solve $2x^2 + 9x + 8 = 0$ correct to 3 s.f by completing the square method. (4 marks)
- (o) The diameter of the taper end of a shaft is measured at two points 30mm apart measured along the surface of the taper. The larger diameter is found to be 12mm and the smaller 6,5mm. Find to the nearest degree the angle between the surface of the taper and its axis. (3 marks)

SECTION BQUESTION 2

- (a) Use the remainder theorem to find the remainder when $2x^3 + x^2 - 7x - 6$ is divided by $(x - 2)$. (4 marks)
- (b) Solve the equation $\log(x - 1) + \log(x + 1) = 2\log(x + 2)$. (8 marks)

QUESTION 3

Given that $y = 4x^2 + 5x - 3$

- (a) Differentiate from the first principle (10 marks)
- (b) Determine the gradient of the curve $y = 4x^2 + 5x - 3$, at $x = -3$. (2 marks)

QUESTION 4

Convert $185,2890625_{10}$ to binary via Octal. (12 marks)

QUESTION 5

The angle of depression of a ship viewed at a particular instant from the top of a 75m vertical cliff is 30° . Find the distance of the ship from the base of the cliff at this instant. The ship is sailing away from the cliff at constant speed and one minute later its angle of depression from the top of the cliff is 20° . Determine the speed in km/hr. (12 marks)

QUESTION 6

- (a) $\int_{-2}^3 (4 - x^2) dx$. (6 marks)
- (b) The velocity of a body, t seconds after a certain instant is $(2t^2 + 5) \text{ m/s}$. Find by integration how far it moves in the interval from $t = 0$ to $t = 4\text{s}$. (6 marks)

QUESTION 7

- (a) Find the surface area of a sphere of a diameter 8cm. (3 marks)
- (b) Find the total surface area of a bucket consisting of an inverted frustum of a cone of slant height 35cm and end diameters of 60cm and 40cm. (5 marks)
- (c) A rectangular football pitch has its length equal to twice its width and a perimeter of 360m. Find its length and width. (4 marks)

QUESTION 8

Determine the equations of the tangent and normal to the curve

$$y = \frac{x^3}{5} \text{ at a point } \left(-1; -\frac{1}{5}\right).$$

(12 marks)

QUESTION 9

Find the inverse of A

$$A = \begin{pmatrix} 5 & 2 & 1 \\ 0 & 6 & 3 \\ 8 & 4 & 7 \end{pmatrix}.$$

(12 marks)

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